Problem 02: Car Washing

Time limit: 1 second

A car washing service has two parallel washing lanes, each consisting of ***N*** sequential cleaning stations. Each station performs a specific washing or detailing task (e.g., rinsing, foaming, scrubbing, waxing, drying).

A car can start at either lane ***1*** or ***2***, and it must pass through all ***N*** stations before leaving. Each station ***(i, j)*** (where ***i*** is the lane number and ***j*** is the station index) has:

- **Processing time *w[i][j]***: The time required to clean the car at station ***j*** on lane ***i***.
- **Switch time *s[i][j]***: The time required to switch the car from lane **i** to the other lane after station ***j***.
- **Entry times *e[1]* and *e[2]*** denote the time to enter lane ***1*** or lane ***2*** at station ***1***.
- **Exit times *x[1]* and *x[2]*** denote the time required to leave after the last station.

The goal is to determine the minimum time required to wash a car completely.

Input Format

The first line contains an integer T $(1 \le T \le 100)$, the number of test cases.

For each test case:

- The first line contains an integer N $(1 < N \le 10^5)$, the number of stations.
- The second line contains two integers $e1, e2$ $(1 \le e1, e2 \le 10^9)$, the entry times for lanes ***1*** and ***2***.
- The third line contains ***N*** integers $w1[1], w1[2], \ldots, w1[N]$ $(1 \le w1[j] \le 10^9)$, the processing times for lane ***1***.
- The fourth line contains ***N*** integers $w2[1], w2[2], \ldots, w2[N]$ $(1 \le w2[j] \le 10^9)$, the processing times for lane ***2***.
- The fifth line contains ***N-1*** integers $s1[1], s1[2], \ldots, s1[N-1]$ $(1 \le s1[j] \le 10^9)$, the switching times from lane ***1*** to lane ***2***.
- The sixth line contains ***N-1*** integers $s2[1], s2[2], \ldots, s2[N-1]$ $(1 \le s2[j] \le 10^9)$, the switching times from lane ***2*** to lane ***1***.
- The seventh line contains two integers $x1, x2$ $(1 \le x1, x2 \le 10^9)$, the exit times for lanes ***1*** and ***2***.

Output Format

For each test case, print a single integer – the minimum time required to completely wash a car.

Sample input	Sample Output
2 5 2 4 7 9 3 4 8 8 5 6 4 5 2 3 1 3 2 1 2 2 3 2 4 10 12 6 7 8 9 5 6 7 8 4 2 3 2 1 3 3 4	32 42